

Piklab: An IDE for PIC Microcontroller

Through this article we explore the features of Piklab over its competitors and its ability to link with several toolchains. Piklab is available for Linux and Windows operating systems on the DVD accompanying this month's issue of EFY Plus

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Piklab is an open source integrated development environment (IDE) for applications devised on Microchip PIC and digital signal processor IC (dsPIC) microcontroller (MCU) that can be compiled only with Qt on Linux and Windows. Qt, being a cross platform framework, allows Piklab to develop non-graphical user interface (non-GUI) programs. Based on general public licence version 2 (GNU GPL v2), Piklab is a complete development environment, providing access to different toolchains.

Piklab, although compatible with all versions of Windows and Linux, is only compatible with K-Desktop Environment 3 (KDE3) and KDE4 from version 0.16.0.

Why use Piklab

Every tool has some specific features that make it essential enough to be used over its competitors. The features that Piklab can be related to are as follows:

1. Provides tools for the complete development process that includes editing and compilation of the source, programming the device and debugging.
2. Integrates with several types of compilers, assembler toolchains and gpusimulators.
3. Supports most common programmers, ICD2 debugger and several types of bootloaders. It also supports some direct programmers for most

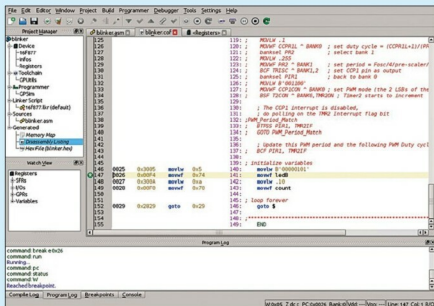


Fig. 1: Debugger

of the devices with PIC18F MCUs and PIC24C EEPROMs.

Other advantages of Piklab

Apart from the key essentials, Piklab has certain other advantages that may be present in other IDEs as well.

Command-line tool.

Piklab is a command-line tool that provides a console for user interface, i.e., it allows the users to issue successive lines of text.

Checks validity of hex files. This tool checks the validity of a hex file to confirm if it complies with the INHX-Standard defined by Intel Corporation,

Associated compilers and assembler toolchains

- GNU PIC utilities and small device C compiler (sdcc)
- PICC compilers (PICC-Lite, PICC and PICC-18)
- MPLAB C18 compiler
- JAL and JALV2 compiler
- BoostC, BoostC++ and BoostBasic compilers
- CCS compiler
- MPC compiler
- CC5X compiler
- MPLAB C30 toolchain

returns the information to the console, cleans the file and fixes the errors. It also compares two hex files.

Supports gpusim. It is a collection of tools from Microchip PIC MCUs that includes gplink, gpasm and gplib. It allows a direct conversion of As-